A Quantitative Approach to Risk Management in Option Selling

Greek-Driven Hedge Strategy on AAPL

FEOF FINANCIAL ENGINEERING OPTIONS FUND

ENGINEERING CONSISTENT PERFORMANCE

Strategy Overview

The Financial Engineering Options Fund (FEOF) is a hedge fund specialized in generating income through option-selling strategies. For this project, we construct a call-writing program on Apple Inc. (AAPL) and manage its risks through Greek-based techniques. The portfolio consists of 10 call options (5 ITM and 5 OTM), with premiums collected upfront and exposure driven by the obligation to potentially deliver shares at maturity.

We evaluate how three hedging approaches—no hedging, Delta hedging, and Delta–Gamma hedging—impact the strategy's profit and loss over time. Using realistic elements such as market-implied volatility, the analysis demonstrates how disciplined Greek-driven hedging can strengthen the stability and performance of a professional option-based income strategy.

Greeks used

1. Delta (Δ): measures how much the option price changes for a small change in the underlying asset price. Is used for delta hedging, the most common form of risk control.

$$\Delta = \frac{\partial U_t}{\partial S_t} = \mathcal{N}(d_1)$$

2. Gamma (Γ): measures how much delta changes as the underlying price changes. High gamma means delta is highly sensitive, therefore, is used for gamma hedging to protect against large market moves.

$$\Gamma = \frac{\partial^2 U_t}{\partial S_t^2} = \frac{\mathcal{N}'(d_1)}{S_t \sigma \sqrt{T-t}}$$

Real Case: No Hedge

At the start of the period, one call option at each strike was sold, generating $135.62 in premiums, which grew to $136.74 by investing it at the risk-free rate. By maturity, AAPL reached $271.40, causing all options to finish in-the-money and requiring a payout of $2,714, while the option deliveries returned only $2,320. After incorporating the premiums and interest, the strategy resulted in a net loss of $257.26.

Real Case: Delta Hedge

Under the Delta Hedge strategy, the same 10 AAPL call options were sold on September 2nd, generating $135.62 in premiums, which grew to $136.74 by investing it at the risk-free rate. Throughout the period, the portfolio's delta was recalculated at each time step, and adjustments were made by buying or selling the underlying whenever the aggregated delta changed.

At maturity, all options expired in-the-money. The exercised positions required a total payout of $2,249.04, while option deliveries returned $2,320. After factoring in premiums, interest, and transaction costs, the Delta Hedge strategy resulted in a net profit of $207.70.

Real Case: Delta-Gama Hedge

The same 10 AAPL call options were sold on September 2nd, generating $135.62 in premiums, which grew to $136.74 at the risk-free rate. The Delta–Gamma strategy expands on Delta hedging by incorporating curvature sensitivity: portfolio gamma was monitored over time, and upper/lower thresholds were calibrated empirically by analyzing how gamma-induced changes in delta would react to price shocks. This ensured that rebalancing occurred only when gamma exposure became large enough to justify adjustment relative to transaction costs.

At each time step, gamma was compared to these thresholds; when breached, the delta hedge was scaled by ±10% to correct curvature-driven drift. With this gamma-adjusted approach, the exercised options required a payout of $1,987.52, while generating proceeds of $2,320. After accounting for premiums, interest, and hedging adjustments, the Delta–Gamma strategy produced a total profit of $469.22.

Monte Carlo Simulation Approach

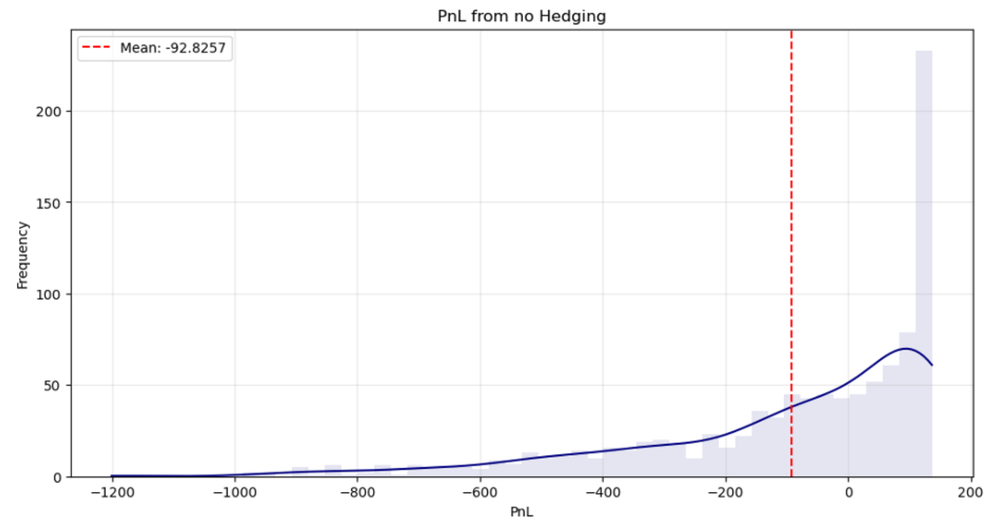
To obtain more robust and statistically reliable results, a set of 1,000 Monte Carlo simulations of the spot price were computed using stock's model:

$$S_t = S_o e^{(\mu_R - \frac{1}{2}\sigma^2)(T-t) + \sigma\sqrt{T-t}W_1}$$

Where:

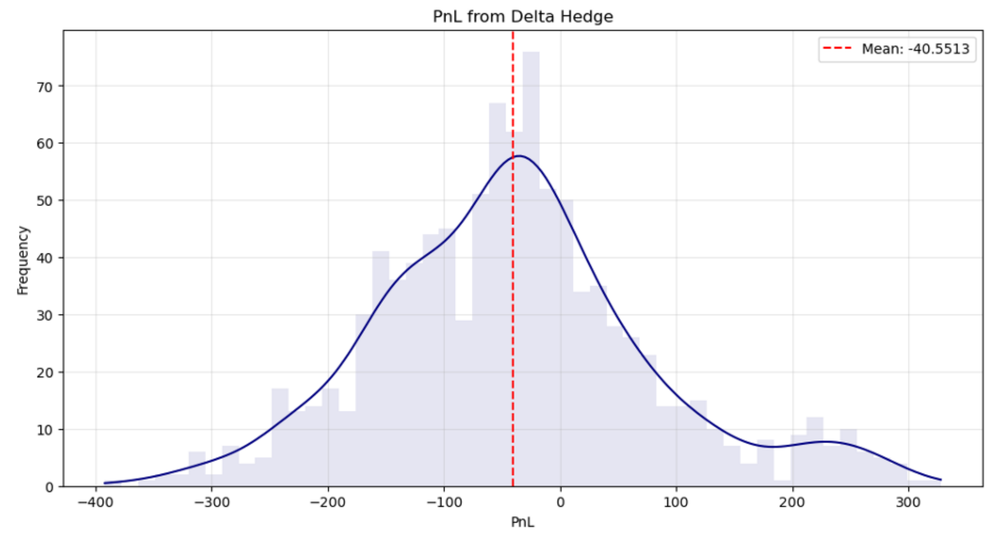
- S_0: actual asset price.
- μ_R: annualized expected return of the asset.
- σ^2: annualized variance of the asset returns.
- $(T-t)$: It's the remaining time until the final horizon T.
- W_1: Wiener process distributed as $\mathcal{N} \sim (0,1)$.

Simulation: No Hedge



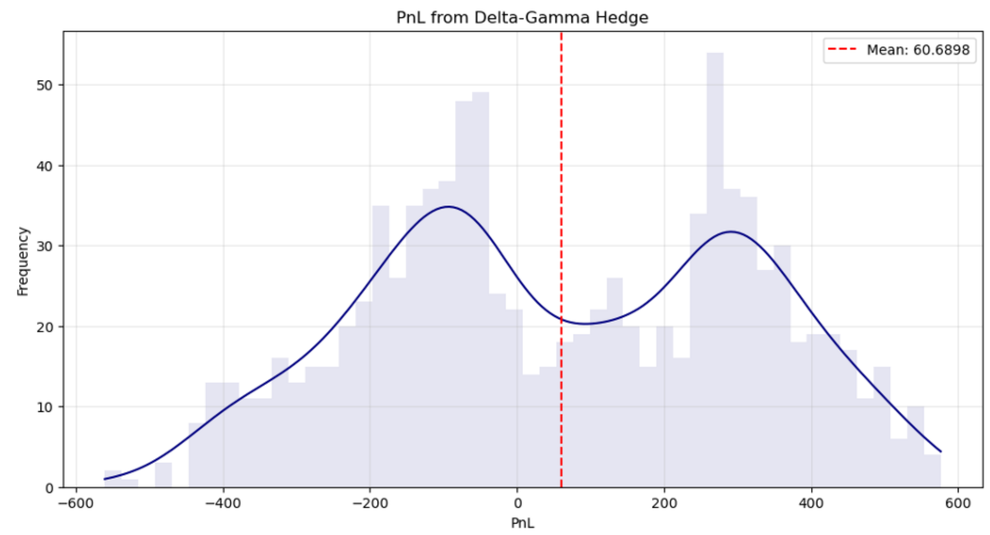
- Probability of losing money without hedging: 52.30%
- 95% VaR without hedging: –$602.49

Simulation: Delta Hedge



- Probability of losing money with Delta hedge: 68.30%
- 95% VaR with Delta hedge: -$226.74

Simulation: Delta-Gamma Hedge



- Probability of losing money with Delta-Gamma hedge: 47.40%
- 95% VaR with Delta-Gamma hedge: -$358.19

Implied Volatility Heatmap

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